Reviewed by: **Professor English**

Comments and Improvements

- One issue that might be worth considering is the political problem with looking into the bank/sovereign nexus. As you say, this could be important in Pakistan, but it could also be uncomfortable, injecting the central bank into fiscal policy and fiscal sustainability issues. Maybe that's something to do internally!

  I agree that it is a sensitive matter. However, the SBP needs to stress test the banks exposure to the government and see how it impacts the solvency of system. At present nothing as such exists.

- Your summary of the current framework in Pakistan was good, but it was also pretty long and left you with a paper that was over the 5 page limit in order to fit in your recommendations. I think you could trim back the summary to fit.

  Incorporated

- Is there really no household sector information in the framework? I would have expected things like home prices and household sector debt?

  Household credit is only 2.1 percent of the total banking sector credit. Due to its small size the SBP does not actively monitor household credit. Mortgage is only 0.5 percent.

- The discussion of NPLs and provisions seemed off to me. At least under US accounting, provisions are supposed to cover expected losses, while capital covers potential losses.

  THE SBPs monitor the provision (NPL) coverage ratio because it is one of the clearest indicators of whether banks have sufficient loan-loss reserves to absorb expected credit losses, thereby preventing those losses from eroding regulatory capital. I have fixed the wording in the paper.

- Is there no composite systemic risk index in Pakistan (p. 6)? I thought that there was a Financial Sector Vulnerability Index? Is that not what you have in mind? But maybe you have in mind developing weights for the composite index based on forecasts of financial crises? More explanation of your recommendation would be helpful.

  We have the FSVI which is published and updated on annual basis for the FSR and nothing like what is published by other central banks. I have made some changes and recommendations to improve the FSVI.

- Which market-based measures in table 3 are available for Pakistan? In part, maybe the the SBP will need to collect some new data?

  Bank level CDS spread and subordinate debt spreads are not available. Information included in the explanation with the table.

- I also like the idea of modeling cross-sector contagion. How would you do that?

  Due to limited space I have included some papers in footnotes for modelling cross-sector contagion.

This paper is my own work. I did not get assistance with this paper from others (beyond Yale writing resources, such as the Yale Writing Center and the SOM Professional Communications Center). I also did not use AI composition software, such as ChatGPT, at any stage of the writing process.

Muhammad Umar Shinwari

19th November 2025

Financial Stability Monitoring – Pakistan

Introduction

Since the GFC, central banks around the world have adopted financial stability monitoring as one of their core responsibilities. This reflects a growing recognition that vulnerabilities can build up in the financial sector even when traditional macroeconomic indicators appear stable. In the case of Pakistan, where the financial sector is bank dominated with 77 percent of the financial sector assets owned by Banks **(Table 1)**, the crucial role of monitoring financial stability is performed by the State Bank of Pakistan (SBP).[1]

Over the years, the SBP has strengthened its monitoring framework through stress testing, consolidated supervision of financial conglomerates, development of a Systemic Risk Survey, broad coverage of financial stability concerns in the Financial Stability Reviews (FSRs) and improvements in supervisory and resolution frameworks. However, despite these developments, the financial stability monitoring system remains largely backward-looking and heavily descriptive with limited integration of forward-looking indicators and composite systemic risk signal that needs further development.[2]

This paper argues that while the SBP's monitoring framework has improved substantially, its effectiveness would be enhanced by integrating forward-looking and market-based indicators, improvements in the composite systemic risk index, strengthening sovereign – bank nexus stress testing and cross sector contagion analysis.

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[1] As per the The SBP Act, Central Bank is entrusted to ensure price and financial stability.
[2] For current The SBP risk index please see: FSVI and Heat Map

Financial stability monitoring at the SBP

Under the amended SBP Act (1956), the State Bank of Pakistan is mandated to safeguard financial system stability and publish an annual Financial Stability Report. Financial stability department leads these assessment by conducting stress testing,[3] consolidated supervision, D-SIB identification, and the compiling financial stability indicators, alongside preparing the Financial Stability Review. To detect emerging vulnerabilities and leverage build-ups, the SBP continuously monitors a broad set of macroeconomic and financial indicators.

Macroeconomic variables: The SBP conducts active surveillance of key macroeconomic indicators including real sector activity, external account, fiscal conditions and inflation dynamics, given their implications for monetary and financial stability. In addition, it also monitors Credit to GDP ratio relative to its long-term trend to assess the buildup of leverage within the financial system **(Chart 1)**.[4]

Capital Ratios: Basel III capital ratios indicate a bank's capacity to absorb losses relative to the riskiness of its assets. The SBP monitors both the Tier 1 Capital to Risk Weighted Assets ratio (RWAs) and the overall Capital Adequacy Ratio (CAR) to ensure banks maintain sufficient high quality capital buffers to remain solvent under stress. Alongside these risk weighted ratios, the SBP also tracks the capital to asset ratio and leverage ratio which specifically captures total exposures including off balance sheet items to assess system wide leverage and balance sheet strength.[5]

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[3] For details on stress testing at THE SBP, please see: Guidelines on Stress Testing.
[4] This measure is recommended by the Basel Committee for setting the Countercyclical Capital Buffer (CCyB).
[5] At present, CAR is set at 11.5 percent whereas LR is set at 3 percent.

Banking Ratios:

i. **Credit Risk:** Credit risk is the most significant risk for banks and requires continuous monitoring. The SBP assesses it through indicators such as the Non-Performing Loans Ratio (NPLR) and the Net NPLs to Net Loans, which reflect stress in lending portfolios. The SBP also monitors provision coverage ratio to ensure that banks maintain sufficient reserves to absorb credit losses without eroding capital. In addition, sector and segment wise NPLs growth is tracked to identify emerging vulnerabilities **(Chart 2).**

ii. **Market Risk:** The SBP monitors equity, exchange rate and interest rate risks on the banks' balance sheets. Equity risk is assessed through banks' equity exposures and related stress tests. Exchange rate risk is monitored via net open positions and stress tests of potential currency depreciation. For interest rate risk, the SBP reviews banks' debt securities portfolio, including duration to gauge sensitivity to rate increases and revaluation losses. It also monitors the gap between rate sensitive assets and liabilities in various maturity buckets.

iii. **Liquidity Risk:** To monitor liquidity risk, the SBP reviews the Basel III Liquidity Coverage Ratio (LCR) for short-term liquidity and Net Stable Funding Ratio (NSFR) for long term funding stability. It also tracks liquid assets to total assets and to deposits to gauge liquidity buffers. Furthermore, the SBP conducts liquidity stress tests using maturity gap analysis and shocks to key indicators to assess banks' resilience under adverse conditions.[6]

Other Variables: In addition to the above risk indicators, the SBP closely monitors banks' net interest margins (NIM) and profitability measures such as return on assets (ROA), return

[6] For details on maturity gap reporting, please see: The SBP Circular 2013

on equity (ROE), and profit after tax (PAT) to assess earnings strength and the capacity to absorb losses. The SBP also compiles a financial sector heatmap **(Table 2)**, to evaluate system-wide and macroeconomic vulnerabilities and to provide forward-looking projections. This heatmap helps identify emerging risks across specific segments of the financial system. Furthermore, the SBP administers a Systemic Risk Survey to captures market participants' perceptions of macroeconomic and financial sector risks, complementing quantitative indicators with expert judgment.[7]

Consolidated Supervision: Consolidated supervision assesses the financial strength and risk profile of an entire group to which a bank belongs, covering risks arising from the bank or from its subsidiaries and affiliates. The SBP monitors consolidated capital ratios, liquidity metrics, intragroup transactions, risk concentrations, and the performance of subsidiaries to assess the overall health of the group.

Gaps in THE SBP Monitoring Framework

Since GFC, financial stability assessment at the SBP has strengthened over the years, however it is still fairly backward looking and has analytical gaps that constrain its early warning capacity. These gaps become more visible when assessed against financial stability monitoring frameworks emphasized in the work of **Adrian, Covitz & Liang (2014)** and **Aikman et al. (2015).**

Firstly, while Pakistan maintains a Financial Stability Vulnerability Index (FSVI), its current design and implementation limit its effectiveness as a real-time systemic-risk monitoring tool. It is produced annually and only published in the FSR, whereas international practice

[7] For details, Please see: THE SBP SRS

emphasizes a higher frequency. Moreover, there is a need to expand the indicator set to incorporate more forward looking metrics **(Table 3)** and applying different statistical weights to indicators based on their forecast of financial crisis

A second major gap is the SBP's **overreliance on backward-looking metrics** such as NPLs ratio, capital and liquidity ratios which capture realized stress but miss the buildup of fragility during calm period. As Adrian et al (2014) note, vulnerabilities often accumulate emphasize that vulnerabilities accumulate when volatility is low and risk appetite is high, making forward- looking market indicators essential for monitoring emerging risks. **Table 4** provides details of possible market indicators that the SBP can use.

A third gap is the **limited quantification of sovereign-bank nexus** in the Financial Stability reviews and stress tests, even though it remain a key vulnerability as 55.5 of the assets are concentrated in government securities[8] and nearly 49 percent of FY25 government revenue went to interest payments **(Chart 3)**. This concentration exposes banks to both market and credit risk in a fiscally constrained environment. However, deeper analysis of this nexus can be politically sensitive, as it touches on fiscal sustainability and may be viewed as the central bank commenting on fiscal policy. As such, while systemically important, fuller assessment may be more suitable for internal surveillance than for public reporting.

Fourth, THE SBP's financial stability monitoring remains **largely siloed across sectors**. While the FSR provides detailed assessments of banks, NBFIs, microfinance institutions and

[8] Source: The SBP FSR 2024

corporates, the SBP does not model cross sector contagion or map interconnectedness across segments of financial sector.[9]

Taken together, these gaps indicate that THE SBP's monitoring approach remains better suited for documenting existing conditions than for anticipating future stress. By embedding forward-looking metrics, integrating sovereign risk, financial-sector cross linkages, and adopting a more holistic assessment of risks, THE SBP could move closer to the kind of vulnerability-based frameworks advocated in recent literature. These reforms would help position Pakistan's financial-stability architecture to respond more effectively to the evolving nature of systemic risk.

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[9] The SBP can draw inspiration from the work done by IMF, Li and Zhang (2024) and Sydow et al. (2024)

References

Aikman, D., Kiley, M., Lee, S. J., Palumbo, M. G., & Warusawitharana, M. (2017). Mapping heat in the US financial system. *Journal of Banking & Finance*, *81*, 36-64.

Adrian, T., Covitz, D., & Liang, N. (2015). Financial stability monitoring. *Annual Review of Financial Economics*, *7*(1), 357-395.

Financial Soundness Indicators SBP

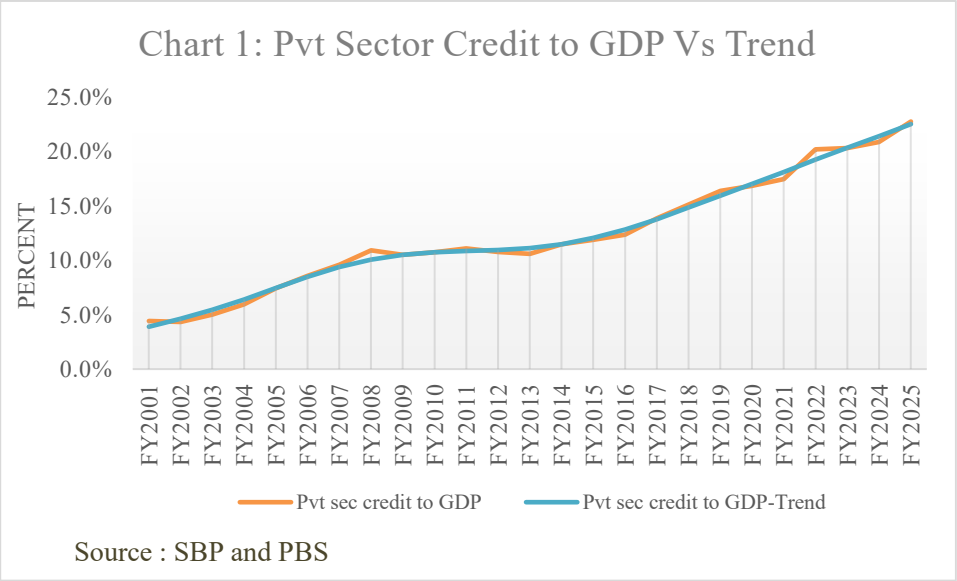
Credit to Private Sector

Pakistan GDP Data

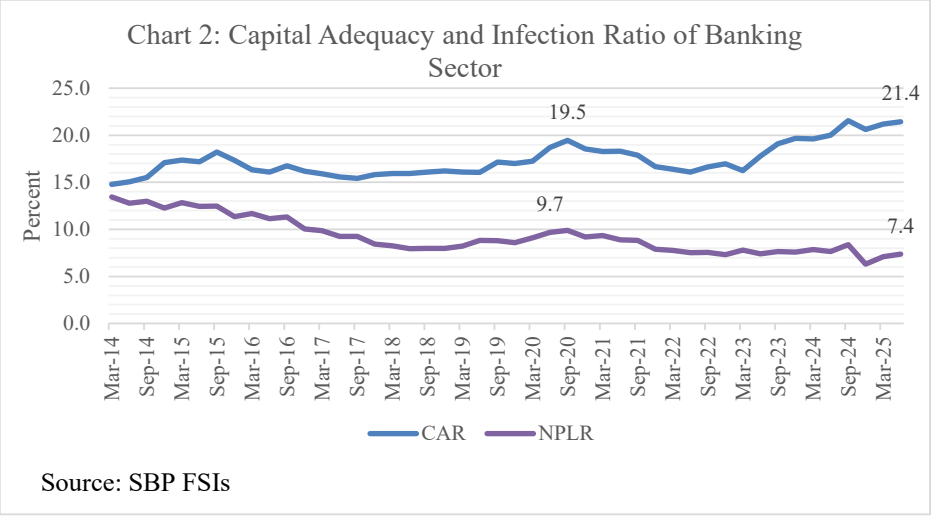
Interest Payments and Revenue, Fiscal Data

Charts and Tables

- Chart below presents the Credit to GDP gap from its trend which has been obtained using Hodrik Prescott filter. Positive gap reflects buildup of leverage in the system.



- While capital ratios are important financial stability indicators they are not forward looking. In case of Pakistan the CAR remained well above the regulatory limit of 11.5 during covid and has been continuously improving since CY23 even though financial conditions were tough. Similarly, NPLR keeps on improving even during times of stress.



- Interest Payments have been a real drag on the government budget due to high level of indebtedness and poor performance on the revenue collection side. Graph below shows the ratio of interest payment to total revenue for Pakistan which reached up to 62 percent during times of high interest rate, which is concerning from financial stability perspective, since banks are highly invested in government securities.

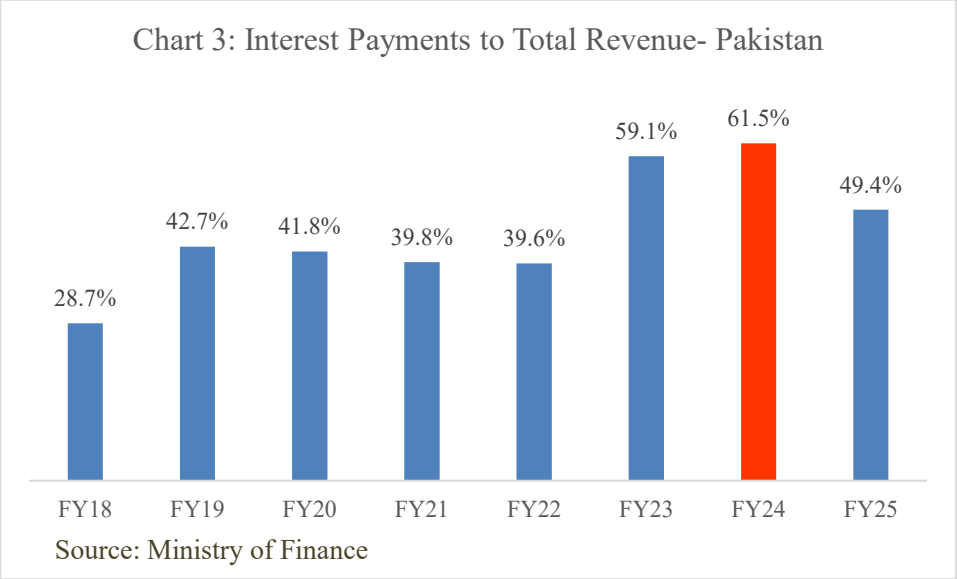
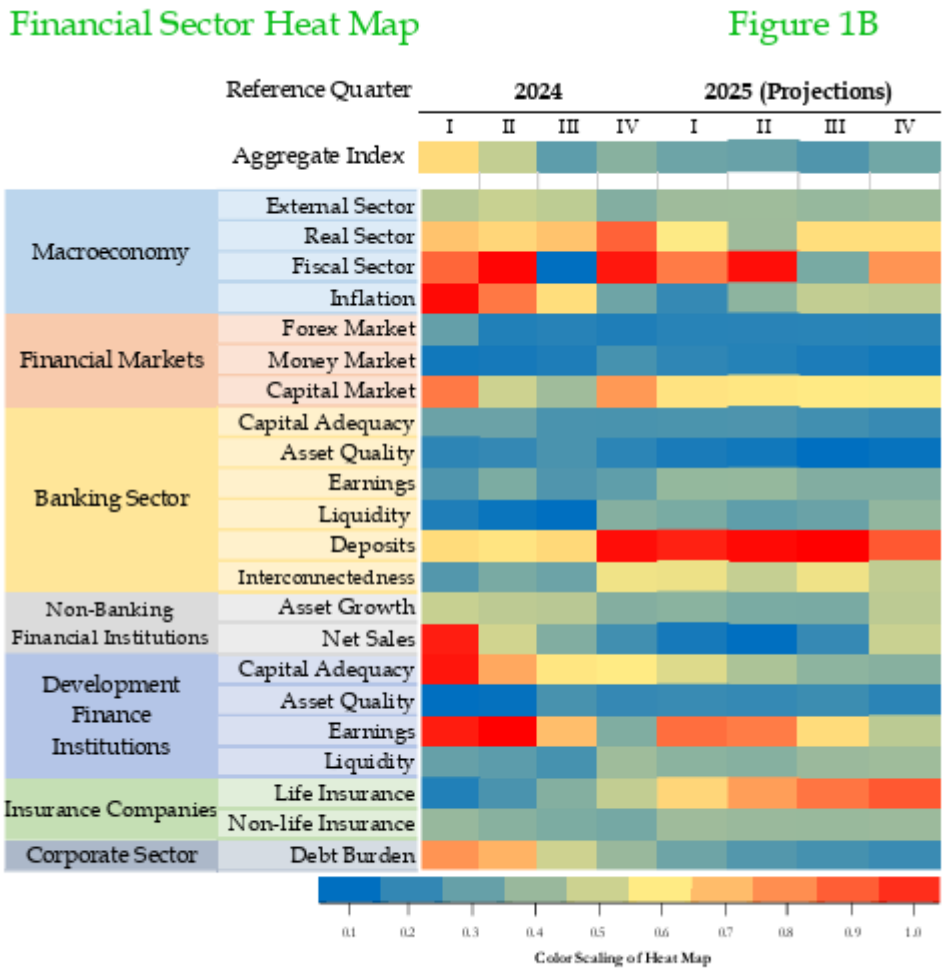


Table 1: Financial Sector Assets

Percent Share in Total Assets			
	2022	**2023**	**2024**
MFBs	1.6	1.3	1.5
DFIs	3.1	4	2.8
NBFIs	5.5	5.8	8.9
Insurance	5.5	5.4	4.9
CDNS	7.3	5.4	4.6
Banks	77	78.5	77.2
Assets as a Percent of GDP			
	2022	**2023**	**2024**
MFBs	1	0.8	1
DFIs	1.9	2.4	1.8
NBFIs	3.4	3.6	5.8
Insurance	3.4	3.1	3.1
CDNS	4.5	3.3	3
Banks	47.8	48.4	50
Overall Assets	62	61.7	64.8

Source: SBP, SECP, CDNS & PBS

Table 2: From THE SBP Financial Stability Review 2024



Source: SBP Staff Estimates

Table below provides details of various indicators used by Aikman et al. (2015) for constructing a risk index. SBP can improve its index by incorporating more market-based indicators, information on NBFIs, leverage transformation and financial sector leverage. A tick mark has been placed in front of indicator that can be worked out in case of Pakistan given data limitations.

Table 3: List of indicators utilized by Aikman et al. (2015) for constructing a systemic risk indicator

Category	Indicator
Risk appetite/Asset valuation	Price to rent ratio Change in price for commercial real estate Spreads on baa and high-yield bonds over treasuries ✓ Issuance of high-yield bonds and leveraged loans Ratio of forward earnings to price for the S&P 500 ✓ Above ratio minus real 10 year Tbill ✓
Non Financial Imbalances	Mortgage debt to GDP relative to trend Debt owned by risky borrowers ✓ Consumer debt to GDP ✓ Consumer debt service to disposable income Consumer debt owned by risk borrowers Gowth rate of NBFI sector debt ✓ NBFI sector ICR ✓ Share of bond issuance by low rated firms Savings of household and businesses net of capital formation
Financial Sector Vulnerabilities	CAR Equity to assets Tier 1 ratio Broker dealer leverage ratio Loans to deposit ratio Maturity gap between assets and liabilities ✓ Net short term wholesale funding relative trend (10Y) at non banks Runnable liabilities such as: fed funds, repos, commercial paper, uninsured deposits, money market mutual funds relative to trend ✓ Ratio of total financial sector liabilities to GDP relative to 10 year moving average ✓ Ratio of top 5 banks assets to total banking sector assets ✓

Table 3: Market-based risk indicators

Indicator*	Literature Reference
Volatility	Sarin & Summers (2016)
Ratio of Bank volatility to market volatility	Sarin & Summers (2016)
CDS Spread of Banks	Sarin & Summers (2016)
Ratio of Bank PE ratio to Market PE ratio	Sarin & Summers (2016)
Market to Book Value	Sarin & Summers (2016)
Subordinate debt spreads	Flannery and Sorescu (1996)
Sovereign CDS Spread	Adrian, Covitz & Liang (2014)
Interest rate swap spreads	Adrian, Covitz & Liang (2014)
Market liquidity indicator (Bid-Ask spreads)	Adrian, Covitz & Liang (2014)

*Housing sector indicators are excluded due to the small size of exposure

Market-based indicator listed below can improve financial stability monitoring at the SBP. From the indicator listed below data on CDS spreads of bank and subordinate debt spreads is not available in Pakistan